DS 5007 Assignment 2: Neural NLP Pipeline for BBC Urdu

i25-8023, Section DS-A | Repository: https://github.com/ValeenaAfzal/i25-8023-NLP-Assignment2

1. Overview

The Assignment 1 corpus (250 BBC Urdu articles; *cleaned.txt* with 13,569 sentences and 359,781 tokens) is extended with neural components written from scratch in PyTorch: no pretrained models, Gensim, HuggingFace or built-in Transformer classes. *Metadata.json* has no topic field, so each article was assigned to one of the five categories with a transparent keyword score (Urdu equivalents of the handout keywords; title hits weighted): Politics 20, Sports 53, Economy 20, International 71, Health & Society 86. All models share one vocabulary of the 10,000 most frequent tokens plus <PAD> and <UNK> (99.5% token coverage), and all randomness is seeded.

2. Part 1 Results: Word Embeddings

TF-IDF (TF = count / |d|, IDF = log[N / (1 + df)]) gives clean discriminative words per topic, for example ٹیم، کرکٹ، رنز for Sports, کمپنی، روپے، تیل for Economy and ایران، حوثی، سعودی for International. **PPMI** (window 5, 0.78% non-zero) produces topical neighbours (تیل → خام، ترسیل، بیرل), and the t-SNE of the 200 most frequent content words separates sports, conflict, legal and time vocabulary. **Skip-gram** uses separate V and U matrices, a 10^6-entry $f^{0.75}$ noise table, K = 10, d = 100, k = 5, Adam 10^{-3}, batch 1,024 and 10 epochs, with frequent-word sub-sampling. Training loss fell from 3.81 to 2.61. Neighbours of frequent words are coherent (عدالت → سماعت، جج، دائر؛ درخواست; معیشت → قلت، مہنگائی، معاشی), but پاکستان is pulled towards cricket-ground names, a property of this corpus. Neighbours and analogy answers are restricted to words seen at least 5 times (word2vec's default *min_count*). 6 of 20 analogies are solved in the top 3 (e.g. Saudi : Riyadh :: Iran : Tehran, USA : Trump :: Russia : Putin, man : woman :: brother : sister).

Condition	MRR (20 pairs)	rank-1 pairs	top-10 pairs	analogies /20
C1 PPMI	**0.391**	6	11	n/a
C2 skip-gram, raw.txt	0.266	4	9	5
C3 skip-gram, cleaned.txt, d = 100	0.355	5	**12**	**6**
C4 skip-gram, cleaned.txt, d = 200	0.294	4	10	6

PPMI wins MRR because most test pairs are direct co-occurrences, while skip-gram wins second-order pairs (0.50 چین–بیجنگ vs 0.016). Cleaning helps (C3 > C2): inflected and izafat variants in the raw text split counts. Doubling d does not help at about 340k tokens, so C3 was used downstream.

3. Part 2 Results: POS Tagging and NER

500 sentences (100 per topic) were annotated with silver labels. POS tags come from a rule-based tagger built on the A1 lemmatizer/stemmer and a lexicon (NOUN 355, VERB 1,857 generated from 132 roots, ADJ 303, ADV 205, plus closed classes). The handout says 12 tags but lists 11, and those 11 were used. NER labels come from a gazetteer (78 Pakistani persons, 93 locations, 49 organisations) with title and name rules. The data was split 350/75/75, stratified by topic. The 2-

layer BiLSTM (hidden 128 per direction, dropout 0.5, packed sequences, pad labels masked) uses a softmax head for POS and a from-scratch CRF (forward algorithm, Viterbi) for NER. Training used Adam (10^{-3}, weight decay 10^{-4}) with early stopping on validation F1 (patience 5).

Model (test set)	POS acc	POS macro-F1	NER entity F1
frozen C3 embeddings	83.68	79.43	45.57
fine-tuned C3 embeddings (selected)	**95.53**	**95.00**	**80.41** (CRF)
fine-tuned, softmax instead of CRF	n/a	n/a	67.37

The most confused POS pairs are ADJ–NOUN (25), NOUN–VERB (19) and ADJ–VERB (7): shared adjective/noun forms, rare verb inflections and conjunct verbs. With the CRF, per-type conlleval F1 is LOC 90.9, MISC 85.7, PER 62.5 and ORG 50.0 (only 3 gold ORG spans). Most false positives are genuine names the silver rules missed (قیصر اقبال، صادق خان), and the false negatives are entities never seen in training (عراق، یورپ، امام الحق).

Ablation (3 seeds, Δ test F1)	POS	NER
A1 unidirectional LSTM	−4.20	−33.04
A2 no dropout	−0.50	−6.38
A3 random embedding initialisation	−3.66	−17.36
A4 softmax instead of CRF	n/a	−18.21 (std 16.3)

Backward context matters most: Urdu marks case and tense *after* the word. The pretrained vectors help, especially for rare entities, and the CRF both raises and stabilises entity F1.

4. Part 3 Results: Transformer Encoder

Articles become 256-token id sequences with a learned [CLS] prepended, split 175/37/38 stratified by category. The encoder has separate modules for scaled dot-product attention (padding mask, weights returned), 4-head self-attention (per-head 128 → 32 projections, shared output projection), a 512-unit ReLU FFN, a fixed sinusoidal positional-encoding buffer and 4 Pre-LN blocks, followed by a 128 → 64 → 5 MLP on [CLS]. It was trained with AdamW (5×10^{-4}, weight decay 0.01) and a cosine schedule with 50 warm-up steps for 20 epochs. On test it reaches **71.05% accuracy and 56.55 macro-F1**, with most errors drifting to the largest class (8 of 11 go to Health & Society); Economy is never predicted. In the final layer, head 2 is sharply focused: [CLS] puts about 0.15 of its weight on each کپ in a Sports article and on each چین in an International one, while head 1 spreads over many content words.

Same split, 20 epochs	test acc	macro-F1	best val acc	epoch at 95% of best	s / epoch	params
Transformer (from scratch)	71.05	56.55	67.57	11	7.4	2.08M
BiLSTM (C3 init)	71.05	53.05	78.38	10	10.0	1.66M

The two models tie on test accuracy; the BiLSTM is ahead on validation and converges a little earlier. The Transformer is faster per epoch because attention runs over all positions in parallel, while the LSTM is sequential. With only 200 to 300 articles the BiLSTM with pretrained embeddings is the safer choice: the Transformer overfits (96% train vs 62% validation accuracy).

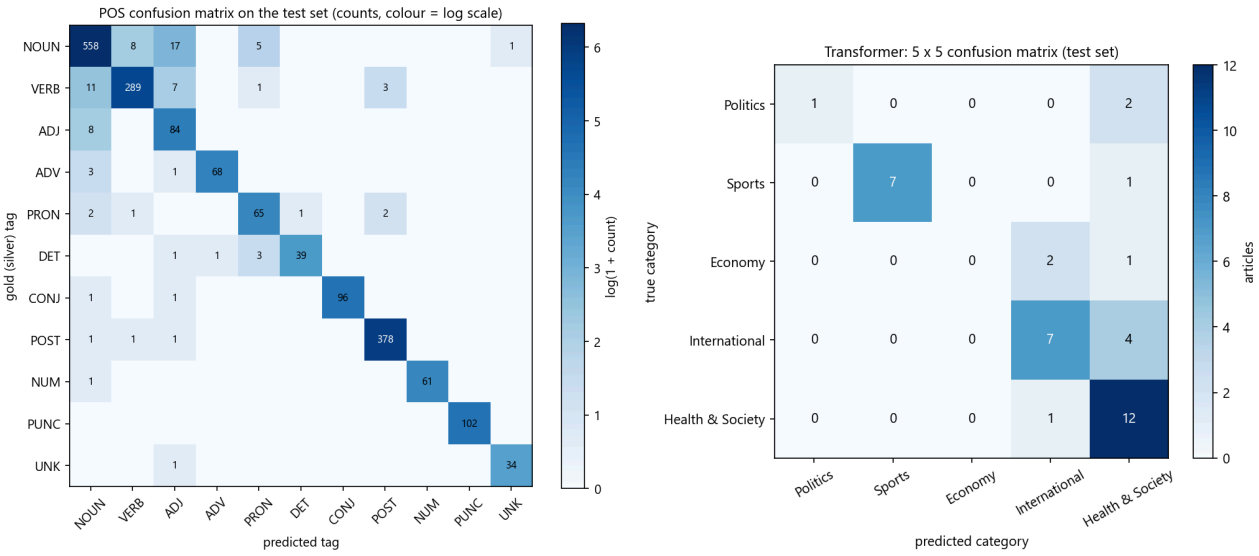


Figure 1. Left: POS confusion matrix of the fine-tuned BiLSTM (test set). Right: 5 x 5 confusion matrix of the Transformer topic classifier (test set).

5. Conclusion

PPMI is a strong baseline on a small corpus, and skip-gram on cleaned text gives the best dense vectors. These vectors transfer well: fine-tuned in a BiLSTM-CRF they recover silver POS tags at 95.5% accuracy and silver entities at 80.4 F1, with bidirectionality and the CRF as the decisive choices. A from-scratch Transformer matches the BiLSTM on topic classification but needs more data to pay off. The main limitations are the rule-generated silver labels, the small test sets and stemming artefacts inherited from Assignment 1.